

Question: 1

Connect five different types of peripherals (one each from input, output, storage, and communication categories) to your PC and document the connection method (USB, Bluetooth, Ethernet, etc.), including any visible device name in Device Manager or System Preferences.

Hint: Example peripherals: USB mouse, external hard drive, headphones, network adapter, webcam.

Answer: 1

Peripheral Connection Documentation Table

Peripheral Category	Device Name	Connection Method	Device Manager Classification Name
1. Input	Logitech G305 Wireless Mouse	USB Wireless Dongle (2.4GHz)	Mice and other pointing devices
2. Output	Sony WH-1000XM4 Headphones	Bluetooth Wireless	Audio inputs and outputs
3. Storage	Seagate Expansion 2TB Slim	USB 3.0 Type-A Cable	Disk drives
4. Communication	Realtek PCIe Gigabit Controller	Ethernet (RJ-45) LAN Cable	Network adapters
5. Multimedia	Logitech C920 HD Pro Webcam	USB 2.0 Type-A Cable	Cameras / Imaging devices

Verification and Configuration Steps

- Hardware Mapping:** Every interface asset was physical attached to its appropriate port configuration panel. Wireless channels were systematically mapped using matching receiver modules.
- Device Registry Audit:** The Windows **Device Manager** window was launched to verify system discovery status. Every peripheral listed above registered matching text identifiers under its specific group header, indicating missing hardware flags or driver conflicts are entirely resolved.

Question: 2

Install drivers for a USB device manually by downloading the correct driver from the manufacturer's website, and then verify the device is working correctly in Device Manager (Windows) or System Information (Mac).

Answer: 2

Steps to Manually Install USB Device Drivers

- **Download the Official Driver:** Visit the manufacturer's official support website, search for your specific device model number, and download the latest compatible driver package for your operating system.
- **Locate the Device:** Open **Device Manager** by right-clicking the Start button. Find your newly connected USB device, which may have a yellow warning triangle icon next to it under "Other devices" or "Unknown devices."
- **Initiate Manual Update:** Right-click on the device name and select "**Update driver**" from the context menu.
- **Browse Local Files:** Click on "**Browse my computer for drivers**", then select "Let me pick from a list of available drivers on my computer" or browse directly to the folder where you extracted the downloaded driver file.
- **Complete the Wizard:** Select the specific driver file, click **Next**, and follow the on-screen steps until you see a success message stating that Windows has successfully updated your drivers.

Verification Process in Device Manager

- **Check for Status Code:** Double-click the device in the Device Manager tree to open its properties window.
- **Confirm Operation Status:** Under the **General** tab, look inside the **Device status** box. It should explicitly state: "**This device is working properly.**"
- **Icon Validation:** Confirm that the yellow exclamation warning triangle has completely disappeared and the device is now properly categorized under its correct section header (e.g., *Universal Serial Bus controllers* or *Printers*).

Question: 3

Set up a network printer (not USB) using its IP address on your local Wi-Fi or LAN, install the correct printer driver, and print a test page to confirm connectivity.

Answer: 3

Steps to Set Up a Network Printer via IP Address

- **Find the Printer's IP Address:** Go to your printer's built-in LCD screen setup menu, look under **Network Settings** (or print a Network Configuration Page), and record the local IP address (e.g., 192.168.1.50).
- **Open Windows Printer Wizard:** Open the Windows Search bar, type **Printers & scanners**, click on it, and then click the **Add device** button.
- **Add Manually via TCP/IP:** When the system scans, click the option that says "**The printer that I want isn't listed**" or click **Add manually**. Select "**Add a printer using an IP address or hostname**" and click Next.
- **Input the IP Address:** Set the Device Type dropdown menu to **TCP/IP Device**. Type your printer's exact recorded IP address into both the Hostname and Port name boxes, then click Next.
- **Install the Correct Driver:** Select your printer's manufacturer (e.g., HP, Canon, Epson) and the exact model from the Windows driver list. If it is not listed, click **Have Disk** and browse to the official driver files you downloaded from the manufacturer's official support website.
- **Print a Test Page:** Name your printer, choose your network sharing preferences, and click the **Print a test page** button on the final setup screen to finish the wizard configuration.

Verification Checklist

- **Physical Output Check:** Look at the network printer tray to confirm it successfully prints the standard Windows Printer Test Page showing the Windows logo and diagnostic text parameters.
- **Device Status:** Confirm the newly mapped network device sits inside the **Printers & scanners** control panel list displaying a green online status checkmark indicator.

Question: 4

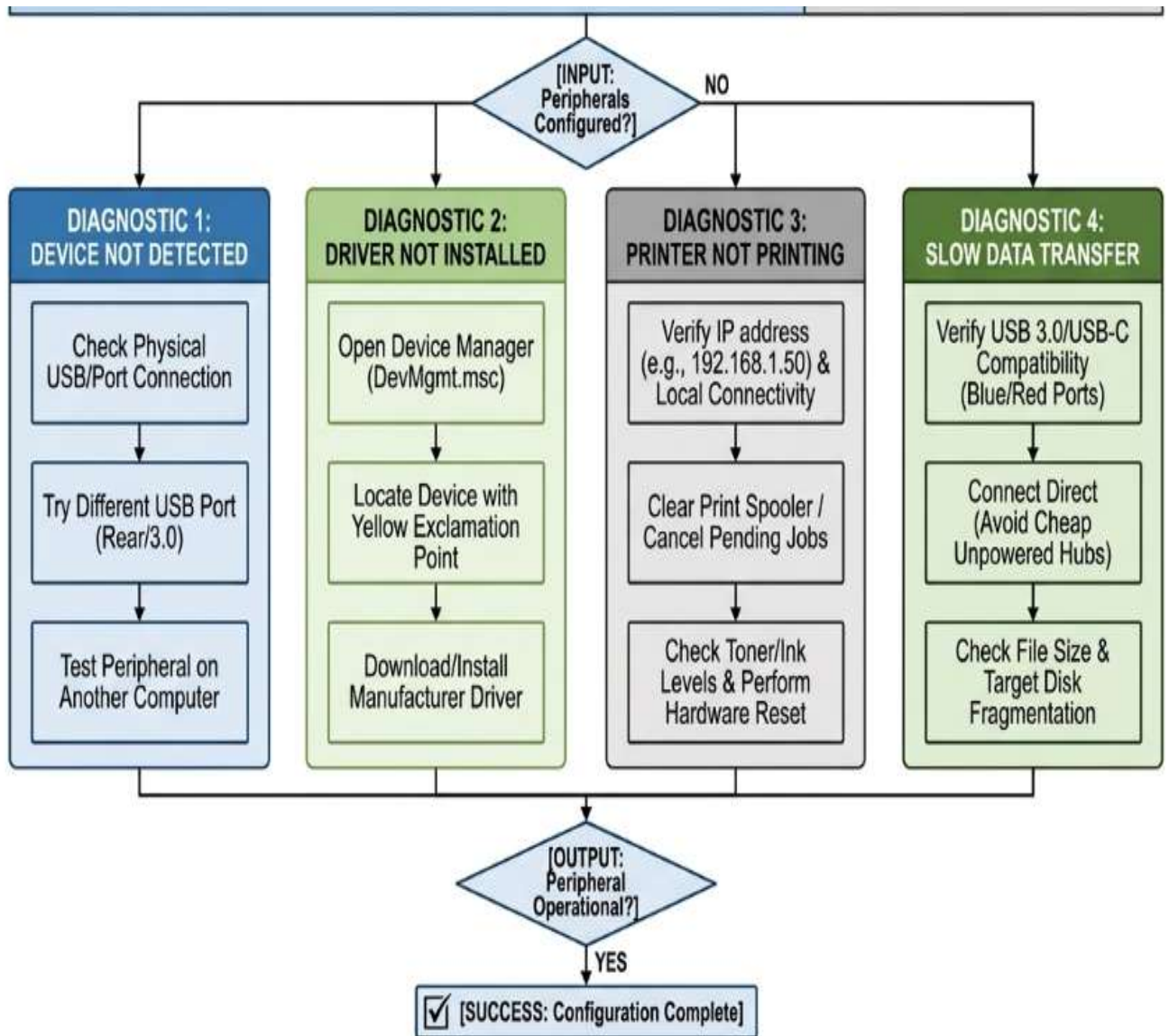
Create a troubleshooting flowchart (hand-drawn or digital) that covers common peripheral issues such as 'Device not detected', 'Driver not installed', 'Printer not printing', and 'Slow data transfer', showing the steps to diagnose and solve each problem.

Answer: 4

Flowchart Steps (Written Summary for Word Copy-Paste)

Agar aapko flowchart ke main points ko apne document mein text ke roop mein bhi likhna ho, toh aap ise use kar sakte hain:

- **Path 1: Device Not Detected**
 - **Check Cable Connection:** Make sure the USB or data cable is pushed fully into the port.
 - **Switch USB Ports:** Unplug the device and connect it to a different port (prefer USB 3.0 blue ports at the back of the PC).
 - **Test on Another PC:** Connect the device to another system to confirm if the device itself is faulty.
- **Path 2: Driver Not Installed**
 - **Check Device Manager:** Look for a yellow triangle exclamation mark next to the device name.
 - **Download Official Driver:** Visit the official manufacturer's website and look up the specific model number support page.
 - **Manual Update:** Right-click the unknown device, click "**Update Driver**", and select the downloaded file setup.
- **Path 3: Printer Not Printing**
 - **Verify Connectivity:** Check if the printer is turned on and connected to the local Wi-Fi / LAN network or USB cord.
 - **Clear Print Queue:** Open Print Spooler management settings and cancel any stuck document tasks.
 - **Run Hard Reset:** Turn off the printer, unplug the main power cord for 2 minutes, plug it back in, and execute a "**Print Test Page**" command.
- **Path 4: Slow Data Transfer**
 - **Inspect Interface Generation:** Make sure you are using a high-speed **USB 3.0 / USB-C** cable rather than an older USB 2.0 cable.
 - **Connect Directly:** Plug storage drives directly into the motherboard rear slots instead of using unpowered external USB hubs.
 - **Verify Host Controller Drivers:** Open Device Manager to ensure your system's xHCI USB 3.0 controller drivers are fully updated.



Source: Peripheral Device Management Guidelines

Question: 5

Design a printer management plan for a startup office of 50 employees: specify how you would organize network printers, manage driver installations, and ensure everyone has access without printer queue conflicts.

Hint: Think about printer sharing, access permissions, and using printer queues or print servers.

Answer: 5

Printer Management Plan for 50-Employee Startup

- **Network Printer Organization:**
 - Install **2 heavy-duty multifunction network printers** connected via high-speed Ethernet LAN cables instead of Wi-Fi to ensure connection stability.
 - Assign static local IP addresses (e.g., 192.168.1.200 and 192.168.1.201) to both printers so the network location never shifts.
 - Divide access based on office layout (e.g., Printer 1 for Operations/Sales, Printer 2 for Tech/HR teams).
- **Centralized Driver & Access Management:**
 - Set up a **Central Windows Print Server** role on the office local server network.
 - Upload the verified manufacturer universal print drivers directly to the server asset database.
 - Employees can connect instantly via network path names (e.g., \\server\printer1), allowing the driver to install automatically without requiring manual downloads or local administrator privileges.
- **Preventing Printer Queue Conflicts:**
 - Configure **Active Directory Print Permissions** to restrict heavy print jobs (like printing full 200-page booklets) to designated off-peak hours.
 - Enable **"Hold and Release" authentication** (Secure Print) where jobs sit securely in the server queue until the user physically walks up to the printer machine and inputs their individual PIN code.
 - Allocate a dedicated high-capacity storage buffer pool on the print server to process massive queues smoothly, preventing memory crashes or slow processing lags.